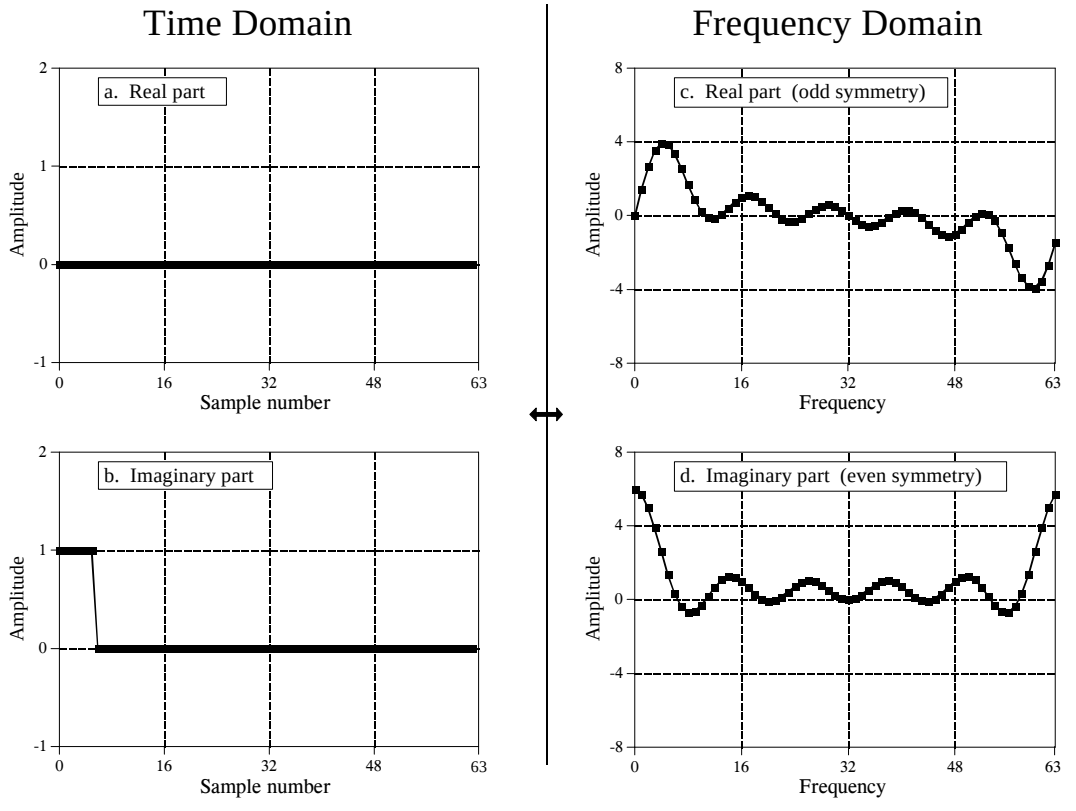

FIGURE 12-11
Imaginary part symmetry of the DFT.

The next step is to modify the algorithm to calculate a *single* DFT faster. It's ugly, but here is how it is done. The input signal is broken in half by using an interlaced decomposition. The $N/2$ even points are placed into the real part of the time domain signal, while the $N/2$ odd points go into the imaginary part. An $N/2$ point FFT is then calculated, requiring about one-half the time as an N point FFT. The resulting frequency domain is then separated by the even/odd decomposition, resulting in the frequency spectra of the two interlaced time domain signals. These two frequency spectra are then combined into a single spectrum, just as in the last synthesis stage of the FFT.

To close this chapter, consider that the *FFT* is to *Digital Signal Processing* what the *transistor* is to *electronics*. It is a foundation of the technology; everyone in the field knows its characteristics and how to use it. However, only a small number of specialists really understand the details of the internal workings.